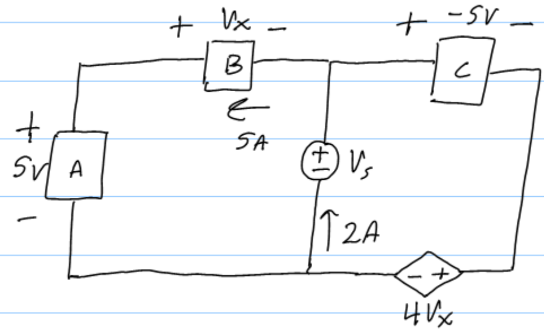
Linear Circuits One (EEL 3004C)

Practice Exam 1 Solutions

Matthew Giannacco

Hunter Volonino



Find V_x, V_s, resistance of A, power of dependent source. Absorbed or delivered?

KVL outer loop:

$$-5 + V_x - 5 + 4V_x = 0$$

$$5V_x = 10$$

$$V_x = 2V$$

KVL left loop:

$$-5 + V_x + V_s = 0$$

$$-5 + 2 + V_s = 0$$

$$-3 + V_s = 0$$

$$V_s = 3V$$

← Substitute V_x →

(Or:) KVL right loop:

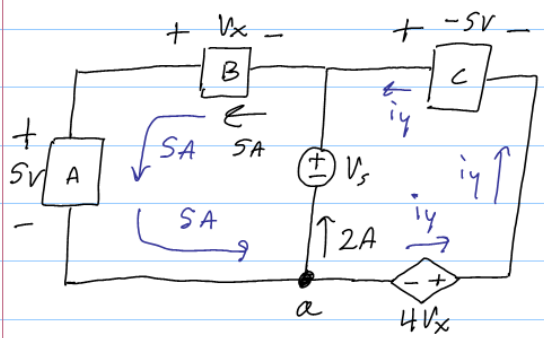
$$-V_s - 5 + 4V_x = 0$$

$$-V_s - 5 + 4(2) = 0$$

$$-V_s + 3 = 0$$

$$V_s = 3V$$

Find resistance of A:



Passive sign, so,

Ohm's Law: $R = \frac{V}{I}$

$$R_A = \frac{5}{5}$$

$$R_A = 1\Omega$$

Find P_{4V_x}:

KCL @ a:

$$-5 + 2 + i_y = 0$$

✱ I have defined i_y here.

$$-3 + i_y = 0$$

$$i_y = 3A$$

Voltage of dependent source:

$$4 \cdot V_x = 4 \cdot 2 = 8V$$

3A, 8V: is not passive sign, so $P = -vi$.

$$P_{4V_x} = -vi = -8 \cdot 3$$

$$P_{4V_x} = -24\,W \text{ (delivering)}$$

$P < 0$, so it is delivering power.

Practice Exam 1 Question 2

Thursday, September 18, 2025 3:11 PM

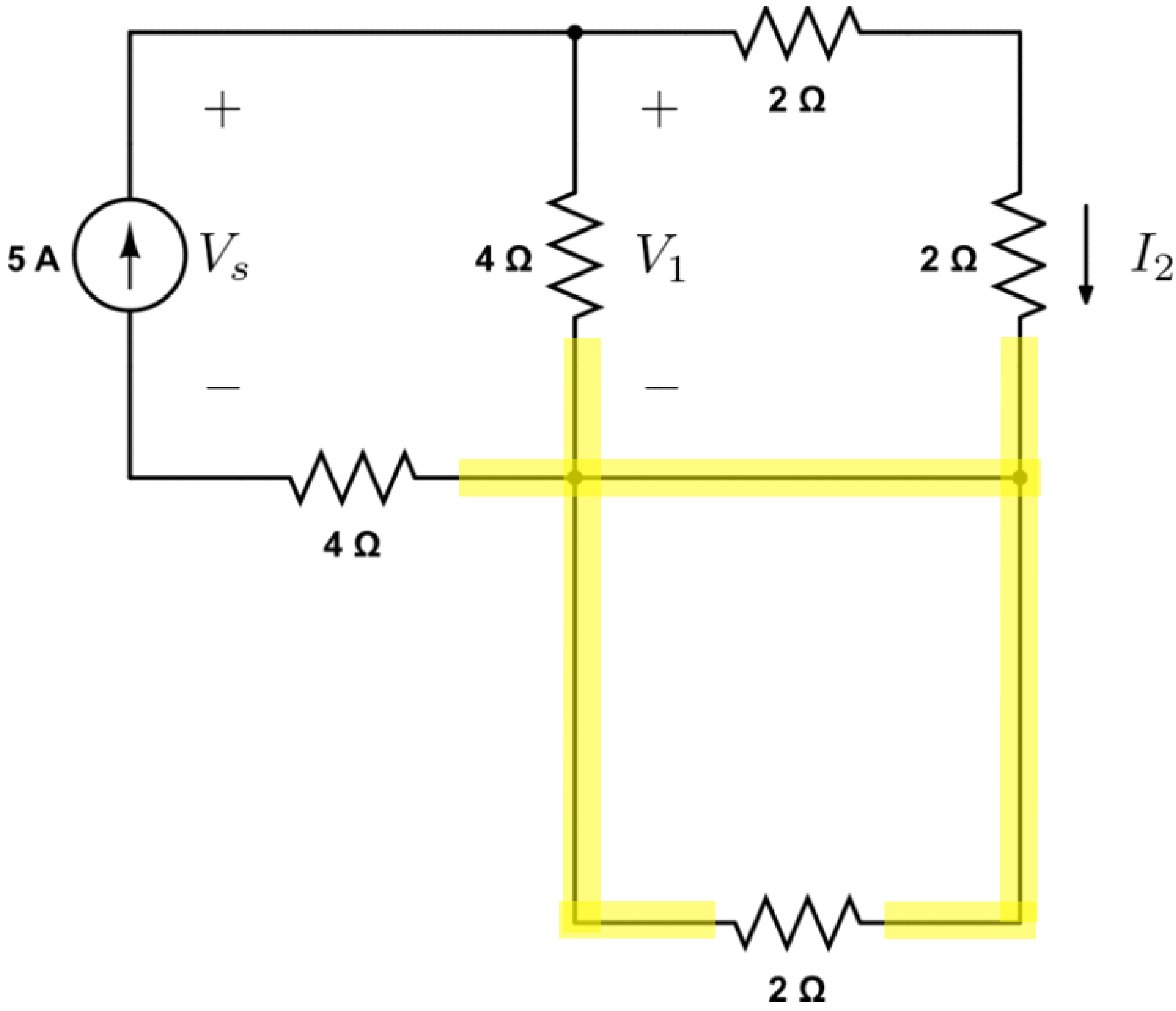
Matthew

For the given circuit, use current division, voltage division, Ohm's Law, and the concept of equivalent resistance to find the following:

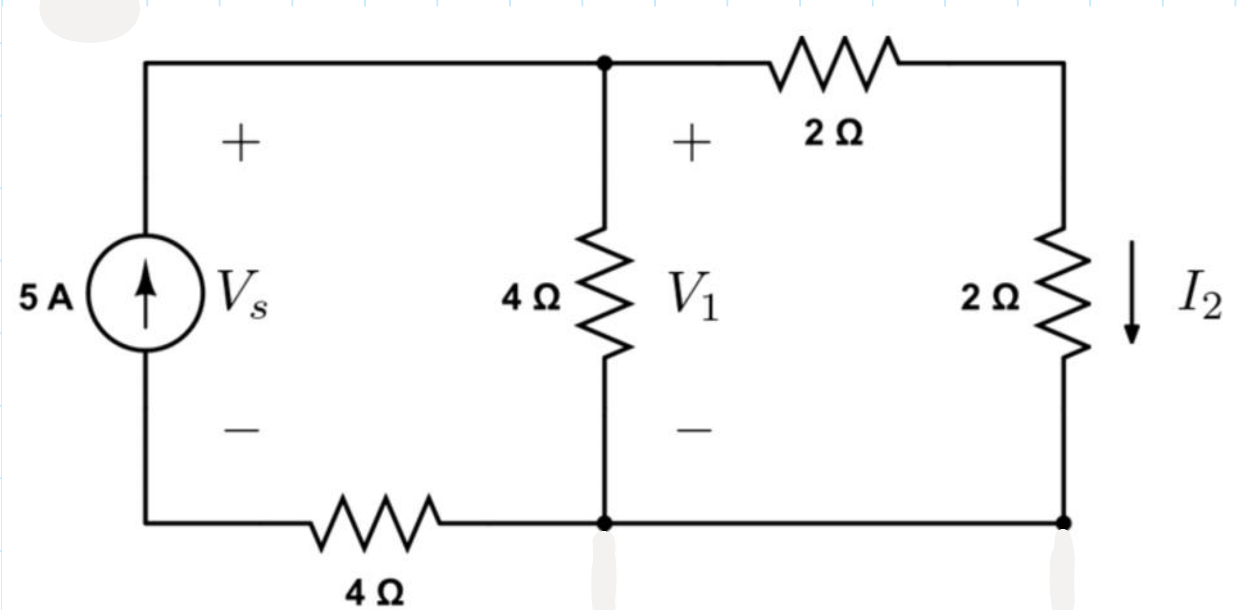
a) V_s

b) V_1

c) I_2



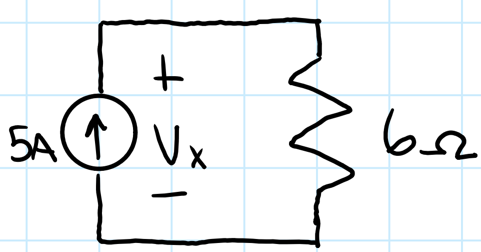
Both sides of the 2Ω resistor are connected to the same node, so it is shorted.



Equivalent Resistance seen by the source:

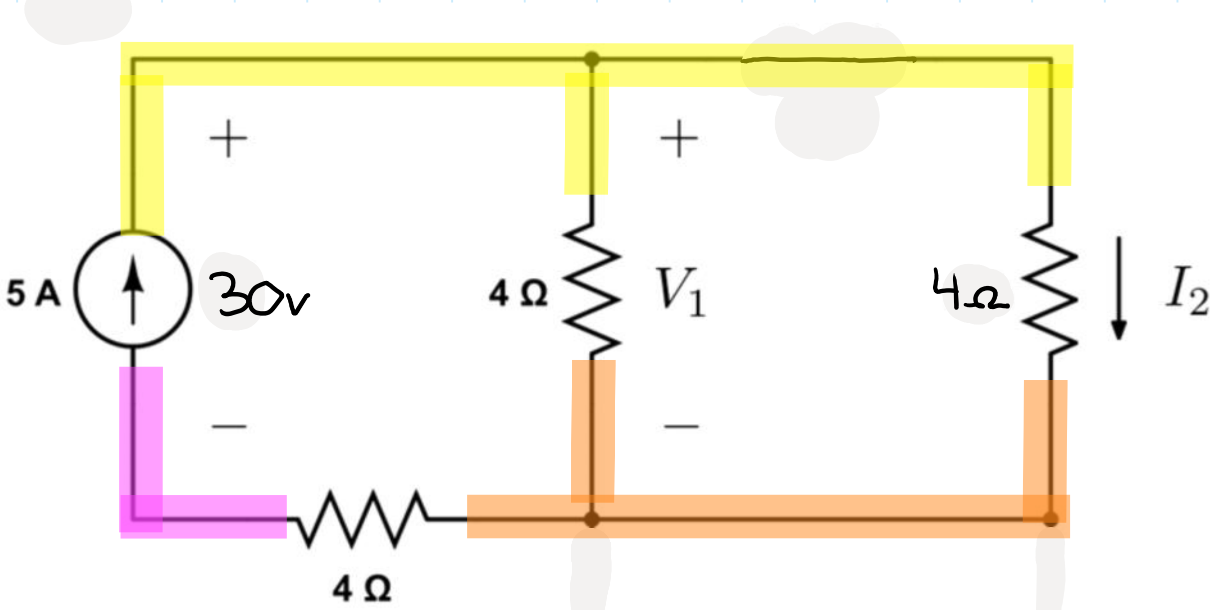
$$\left((2\Omega + 2\Omega) \| 4\Omega\right) + 4\Omega = 6\Omega$$

Equivalent Circuit:



KVL: $-V_x + 6(5A) = 0$

$V_x = 30v$



We can find V_1 with Voltage Division

The right and middle resistor share V_1 volts, but $V_1 \neq 30v$

$$V_1 = \frac{(4\Omega \| 4\Omega)}{6\Omega} \cdot 30v = 10v$$

We can find I_2 with Current Division

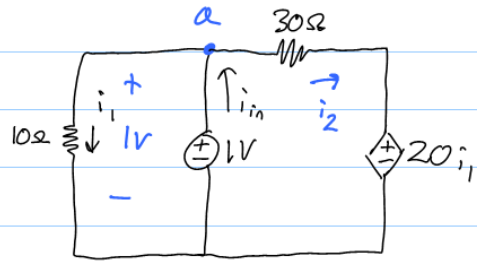
$$I_2 = \frac{(4\Omega \| 4\Omega)}{4\Omega} \cdot 5A = 2.5A$$

Equivalent Resistance for Current Division
is for parallel resistors only

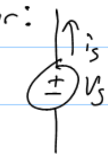
Equivalent Resistance for Voltage Division
is for series resistors only

$V_s = 30v$ $V_1 = 10v$ $I_2 = 2.5A$

Note Title 9/18/2025



Find R_{eq} seen by 1V source.

For: 

$R_{eq} = \frac{V_s}{i_s}$. $V_s = 1$, so $R_{eq} = \frac{1}{i_s}$.

So, $\boxed{R_{eq} = \frac{1}{i_{in}}}$.

10Ω resistor is in parallel with 1V source.

$$i_1 = \frac{V}{R} = \frac{1}{10} A = 0.1A$$

KVL right loop:

$$-1 + 30i_2 + 20i_1 = 0$$
$$-1 + 30i_2 + 20(0.1) = 0$$
$$-1 + 30i_2 + 2 = 0$$
$$30i_2 = -1$$
$$i_2 = -\frac{1}{30} A$$

KCL @ a:

$$i_1 - i_{in} + i_2 = 0$$
$$\frac{1}{10} - i_{in} + \left(-\frac{1}{30}\right) = 0$$
$$\frac{1}{15} - i_{in} = 0$$
$$i_{in} = \frac{1}{15} A$$

$$R_{eq} = \frac{1}{i_{in}} = \frac{1}{\frac{1}{15}} \Omega.$$

$$R_{eq} = 15\Omega.$$